Mutually exclusive events

1 Is it possible for the following two events to happen at the same time?

- It is a Friday.
- It is raining.

2 A die is rolled. Is it possible for the following two outcomes to happen at the same time?

- Rolling a 6.
- Rolling an odd number.

3 A coin is tossed. Is it possible for the following two outcomes to happen at the same time?

- The coin lands on heads.
- The coin lands on tails.

4 Given a standard deck of cards, determine which pair of events cannot happen together:

- Selecting a heart.
- Selecting a 5.
- Selecting a queen.
- Selecting a red card.

5 A six-sided die is rolled. Determine which pair of events can happen together:

- Rolling a number less than 3.
- Rolling a 3.
- Rolling a 5.
- Rolling a multiple of 3.

6 When picking a random card from a standard pack, which two of the four events listed share no common outcomes?

- **Event A:** picking a black card
- **Event B**: picking a King
- **Event C:** picking a spade
- **Event D**: picking a club

7 If A and B are mutually exclusive and $P\left(A\right) = \dfrac{1}{4}$ and $P\left(B\right) = \dfrac{1}{2}$, find $P\left(A \cup B\right)$.

8 If A and B are mutually exclusive and $P\left(A\right) = \dfrac{1}{8}$ and $P\left(B\right) = \dfrac{7}{8}$, find $P\left(A \text{ or } B\right)$.

9 Two events A and B are mutually exclusive. If $P\left(A\right) = 0.37$ and $P\left(A \text{ or } B\right) = 0.73$, find

10 A random card is picked from a standard deck. Find the probability that the card is:

a A spade or diamond.

b A king or a 3.

c An ace of spades or a king of hearts.

Non-mutually exclusive events

11 If $P(A) = \frac{1}{5}$, $P(B) = \frac{1}{6}$ and $P(A \cap B) = \frac{1}{30}$, find $P(A \cup B)$.

12 If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{5}$ and $P(A \cup B) = \frac{2}{15}$, find $P(A \cap B)$.

13 If $P(A \cup B) = \frac{7}{15}$, $P(A \cap B) = \frac{1}{15}$ and $P(A) = \frac{1}{3}$, find $P(B)$.

14 If $P(A) = 0.8$, $P(B) = 0.75$ and $P(A \text{ and } B) = 0.6$, find $P(A \text{ or } B)$.

15 If $P(A) = \frac{1}{7}$, $P(B) = \frac{1}{2}$ and $P(A \text{ and } B) = \frac{1}{14}$, find $P(A \text{ or } B)$.

16 If $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{4}$ and $P(A \text{ or } B) = \frac{1}{3}$, find $P(A \text{ and } B)$.

17 A random card is picked from a standard deck. Find the probability that the card is:

a Red or a diamond.

b An ace or a diamond.

c An ace of spades or an ace of clubs.

d A black or a face card.

18 A standard six-sided dice is rolled. Find the probability of rolling:

a The number 4.

b An odd number.

c A 4 or an odd number.

d An even or prime number.

e A number that is even and prime.

f An even number or a factor of 6.

g A factor of 9 or an even number.

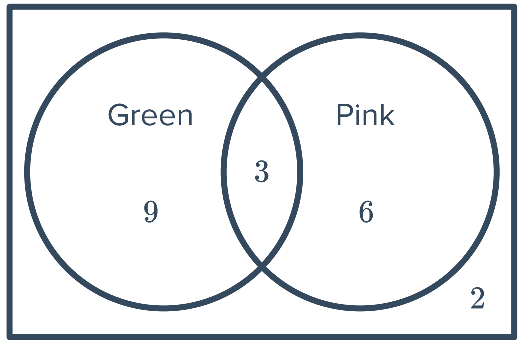
h A number less than 3.

i A number greater than 5.

j a number less than 3 or greater than 5.

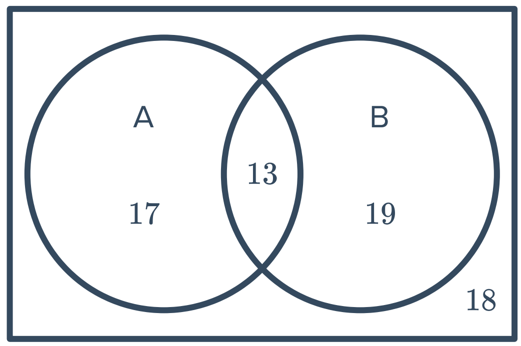
19 The Venn diagram shows the results of a

children liked:
Find the probability of a child liking:



- **a** Green and pink.
- **b** Green or pink but not both.

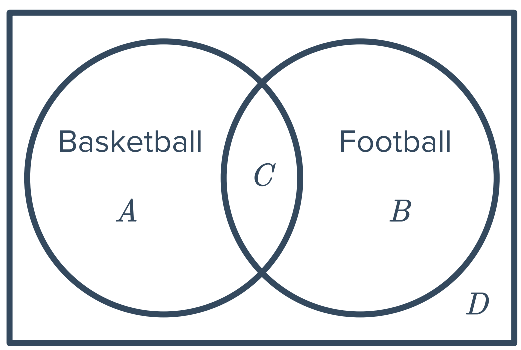
20 Consider the Venn diagram:
Find:



- **a** $P(A)$
- **b** $P(B)$
- **c** $P(\text{not } A)$
- **d** $P(\text{not } B)$
- **e** $P(B \text{ only})$

21 Out of 23 school kids, 12 play basketball and 13 play football, whilst 5 play both sports.

- **a** For the given Venn diagram, find the value of:
  - **i** A **ii** B **iii** C **iv** D



- **b** Find the probability that a student plays football or basketball, but not both.
- **c** Find the probability that a student plays both football and basketball.

22 In a music school of 129 students, 83 students play the piano, 80 students play the guitar and 14 students play neither.

- **a** Construct a Venn diagram for this situation.
- **b** Find the probability that a student chosen at random plays:
  - **i** Both the piano and the guitar.
  - **ii** The piano or the guitar.
  - **iii** Neither the piano nor the guitar.